Cycle Gravity Spectra and a Uniform Counterexample Family for Graffiti 197

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Abstract. Roucairol and Cazenave exhibited the cycle on 17 vertices as a counterexample to Graffiti Conjecture 197 and conjectured that every cycle of order $4r + 1 \geq 17$ is likewise a counterexample, under the usual spectral-range convention. We establish a stronger uniform conclusion: every cycle C_n with $n \geq 15$ violates the proposed inequality. The proof decomposes a shifted reciprocal-distance matrix into an explicit positive-semidefinite sum of cyclic interval Gram matrices, yielding the uniform bound $\lambda_{\min}(R_n) \geq -3/2$. A harmonic-number comparison completes both parity cases without asymptotics. Exact rational interval certificates verify that the threshold is sharp within the class of cycles: $C_3, \ldots, C_{14}$ do not violate the inequality. We supply an independently executable verifier, including exact finite checks and independent Fourier spectral tests. Attribution and historical priority are separated from mathematical correctness pending independent review.

Keywords. spectral graph theory; Graffiti conjectures; reciprocal-distance matrix; circulant matrix; positive semidefinite decomposition; certified computation.

1 The problem and its conventions

The Graffiti programme proposed many inequalities connecting graph invariants; their formulation and automated falsification have a long history [1, 2, 3]. Roucairol and Cazenave [5, §5.1] reported C_{17} as a counterexample to Graffiti 197, and also checked C_{21} and C_{25}. They conjectured that every C_{4r+1} of order at least 17 is a counterexample. Their discussion emphasizes a genuine terminological ambiguity: *range* can mean a numerical span or the number of distinct values. Our claims use *only* the numerical span, as in their counterexample.

Let C_n ($n \geq 3$) have vertex set $\mathbb{Z}/n\mathbb{Z}$, ordinary graph distance $d(u, v) = \min(|u - v|, n - |u - v|)$, and adjacency matrix A_n. If $\lambda_1(A_n) \geq \cdots \geq \lambda_n(A_n)$, write $\lambda_{n-1}(A_n)$ for its second-smallest eigenvalue. Following the gravity convention reproduced in [5, §5.2], every cycle vertex has degree two and hence

$$(\mathrm{Gr}_n)_{uv} = \begin{cases} 0 & u = v, \\ \dfrac{4}{(n-1)d(u,v)} & u \neq v. \end{cases} \tag{1}$$

Let R_n denote the reciprocal-distance matrix, so $\mathrm{Gr}_n = 4R_n/(n-1)$. Define $\mathrm{spr}(M) = \lambda_{\max}(M) - \lambda_{\min}(M)$ for symmetric M. In this convention Graffiti 197 asks for

$$-\lambda_{n-1}(A_n) \leq \mathrm{spr}(\mathrm{Gr}_n). \tag{2}$$

We show that its strict reverse holds on every sufficiently large cycle, with an exact threshold.

Theorem 1 (Infinite counterexample family)**.** *For every integer $n \geq 15$,*

$$-\lambda_{n-1}(A_n) > \mathrm{spr}(\mathrm{Gr}_n). \tag{3}$$

Thus the Roucairol–Cazenave conjecture holds, with a stronger conclusion covering all residue classes of n.

The spectral estimate driving the proof concerns the Harary or reciprocal-distance matrix, an established graph invariant [4].

2 A positive-semidefinite identity

Let $m = \lfloor n/2 \rfloor$, let J be the all-ones matrix, and, for $2 \leq j \leq m$, let B_j be the $n \times n$ binary circulant matrix whose vth column is the indicator of the cyclic interval $\{v, v + 1, \ldots, v + j - 1\}$. Set

$$w_j = \begin{cases} \dfrac{2}{(j-1)j(j+1)} & 2 \leq j < m, \\ \dfrac{1}{m(m-1)} & j = m. \end{cases} \tag{4}$$

All these coefficients are strictly positive.

Lemma 1 (Exact Gram decomposition)**.** *For every $n \geq 4$,*

$$R_n + \frac{3}{2}I = \frac{1}{m}J + \sum_{j=2}^{m} w_j B_j B_j^{\mathsf{T}}. \tag{5}$$

In particular $\lambda_{\min}(R_n) \geq -3/2$.

Proof. Two cyclic intervals of length $j \leq m$ at distance $d \leq m$ share precisely $(j - d)_+ = \max(j - d, 0)$ vertices. Hence the (u, v) entry of $B_j B_j^{\mathsf{T}}$ equals $(j - d(u, v))_+$. For $1 \leq d \leq m$, the discrete convex interpolation of $f_d = 1/d - 1/m$ satisfies

$$\frac{1}{d} - \frac{1}{m} = \sum_{j=d+1}^{m} w_j (j - d),$$

with the coefficients in (4): for $j < m$ they are $1/(j-1) - 2/j + 1/(j+1)$, and the terminal coefficient is $1/(m - 1) - 1/m$. Thus the off-diagonal entries in (5) equal $1/d$.

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On the diagonal,

$$\frac{1}{m}+\sum_{j=2}^{m} jw_j = \frac{1}{m}+\sum_{j=2}^{m-1}\left(\frac{1}{j-1}-\frac{1}{j+1}\right)+\frac{1}{m-1} = \frac{3}{2},$$

including the case $m = 2$ (empty middle sum). This proves the matrix identity. Both J and each $B_jB_j^{\mathsf{T}}$ are positive semidefinite, giving the eigenvalue bound. □

The identity is finite-dimensional and exact; its verification requires no numerical eigensolver. Let $H_m = \sum_{j=1}^{m} j^{-1}$. As R_n is nonnegative with constant row sums, its largest eigenvalue is that row sum. In particular,

$$\lambda_{\max}(R_n) = \begin{cases} 2H_m & n = 2m+1, \\ 2H_{m-1}+1/m & n = 2m. \end{cases} \tag{6}$$

Combining (1), Lemma 1 and (6) yields

$$\operatorname{spr}(\mathrm{Gr}_n) \le \frac{4}{n-1}\left(\lambda_{\max}(R_n)+\frac{3}{2}\right). \tag{7}$$

3 Proof of the universal statement

For cycles the adjacency spectrum is exactly

$$\{2\cos(2\pi k/n) : k = 0, 1, \dots, n-1\}.$$

Ordering these eigenvalues, with multiplicity, gives

$$-\lambda_{n-1}(A_n) = \begin{cases} 2\cos(\pi/(2m+1)) & n = 2m+1, \\ 2\cos(\pi/m) & n = 2m. \end{cases} \tag{8}$$

Proof of Theorem 1. For odd $n = 2m+1$ ($m \ge 7$), (7) bounds the gravity spread by

$$q_o(m) = \frac{4H_m+3}{m}.$$

This is strictly decreasing: $q_o(m) > q_o(m+1)$ follows from $4H_m + 3 > 4m/(m+1)$. Meanwhile $2\cos(\pi/(2m+1))$ is increasing in m. It therefore suffices to check $m = 7$. Using $H_7 = 363/140$, $\cos x \ge 1 - x^2/2$, and the rational inequality $\pi < 22/7$, we get

$$2\cos(\pi/15) - q_o(7) > \left[2-\left(\frac{22}{105}\right)^2\right] - \frac{468}{245} = \frac{506}{11025} > 0.$$

This proves the odd case.

For even $n = 2m$ ($m \ge 8$), define

$$S_m = 2H_{m-1}+\frac{1}{m}+\frac{3}{2}, \qquad q_e(m) = \frac{4S_m}{2m-1}.$$

Now $S_{m+1} - S_m = 1/m + 1/(m+1)$; thus $(2m-1)(S_{m+1} - S_m) < 4 < 2S_m$ for $m \ge 2$, proving $q_e(m) > q_e(m+1)$. The quantity $2\cos(\pi/m)$ increases. At $m = 8$,

$$2\cos(\pi/8) - q_e(8) > \left[2-\left(\frac{11}{28}\right)^2\right] - \frac{1907}{1050} = \frac{1733}{58800} > 0.$$

Both estimates compare the left-hand side of (3) to an *upper bound* on its right-hand side. The strict inequality follows for all $n \ge 15$. □

Corollary 1. *For every $r \ge 4$, the cycle C_{4r+1} refutes Graffiti 197 in the numerical-span convention, as conjectured in [5].*

4 Sharpness and certificate design

To check the finite complement without trusting floating-point eigenvalues, use a Fourier eigenvalue of R_n at frequency $k = \lfloor n/2 \rfloor$:

$$\eta_k = 2\sum_{d=1}^{\lfloor (n-1)/2 \rfloor} \frac{\cos(2\pi kd/n)}{d} + \mathbf{1}_{2\mid n}\frac{(-1)^k}{n/2}. \tag{9}$$

We have $\lambda_{\min}(R_n) \le \eta_k$, regardless of whether this mode is globally minimal. It follows that

$$-\lambda_{n-1}(A_n) - \operatorname{spr}(\mathrm{Gr}_n) \le 2\cos\theta_n - \frac{4}{n-1}(s_n - \eta_k), \tag{10}$$

where $s_n = \lambda_{\max}(R_n)$ and $\theta_n = \pi/n$ for odd n, $2\pi/n$ for even n.

The accompanying Python verifier encloses π by exact rational alternating-series estimates from Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. Every cosine is enclosed by exact rational evaluations of its Taylor polynomial of degree 24, with the explicit remainder bound $(22/7)^{25}/25!$. It then checks that the *upper* bound on the right-hand side of (10) is strictly negative for each $n = 3, \dots, 14$. All acceptance-path comparisons are between exact Python rational numbers. Thus none of these twelve smaller cycles refutes (2).

Theorem 2 (Sharp cycle threshold, computer-assisted finite part)**.** *Among simple cycle graphs C_n ($n \ge 3$), violation of (2) occurs if and only if $n \ge 15$. The positive direction has the analytic proof above; the finite converse is certified by the rational interval algorithm just described.*

5 Formal certification: progressive twelve-stage proof ledger

A pinned Lean 4/mathlib v4.24.0 project certifies proof-critical identities of Lemma 1; sources and lockfile reside in the public Newton Astro Labs repository. The full graph-spectral theorem is *not yet kernel-certified.*

Stage 1: rational margins and Gram positivity. `NewtonConjectures.lean` certifies the strict base-case margins 506/11025 and 1733/58800, rational reciprocal second differences, positive weights, and nonnegative weighted Gram quadratic forms.

Stage 2: universal reciprocal-distance kernel. `CycleKernel.lean` proves telescoping for all $f : \mathbb{N} \to \mathbb{Q}$. For $1 \le d < m$, the reciprocal specialization is

$$\frac{1}{d} = \frac{1}{m} + \sum_{j=d+1}^{m-1}(j-d)\frac{2}{(j-1)j(j+1)} + \frac{m-d}{m(m-1)}. \tag{11}$$

Stage 3: exact cyclic Gram entries. `CyclicIntersection.lean`, `CyclicZMod.lean` and `CyclicGram.lean` formalize genuine cyclic incidence matrices over $\mathbb{Z}/n\mathbb{Z}$ and their `Matrix.mul`. For all vertices u, v and $2j \le n$, including wraparound, disjoint arcs, and antipodes, the kernel proves

$$(B_j B_j^{\mathsf{T}})_{uv} = \max\{j - d(u,v), 0\}. \tag{12}$$

Stage 4: actual weighted Gram diagonal. For every $m \ge 2$, `GramDiagonal.lean` certifies

$$\frac{1}{m} + \sum_{j=2}^{m-1} \frac{2}{(j-1)(j+1)} + \frac{1}{m-1} = \frac{3}{2}. \tag{13}$$

It also proves the identical diagonal for the *actual weighted incidence matrices* of every C_n, $n \ge 4$.

Stage 5: all off-diagonal distances. For every $m \ge 2$ and $1 \le d \le m$, `OffDiagonal.lean` proves

$$\frac{1}{m} + \sum_{j=2}^{m} w_j \max\{j - d, 0\} = \frac{1}{d}. \tag{14}$$

These are exactly the weights of (4), with the terminal case $d = m$ included using natural truncated subtraction.

Stage 6: the complete rational matrix equality. The new kernel-checked module `FullGram.lean` formally defines the shifted reciprocal-distance matrix over $\mathbb{Z}/n\mathbb{Z}$, with diagonal $3/2$ and off-diagonal entries $1/d(u,v)$. For every integer $n \ge 4$, the theorem `cyclic_shifted_reciprocal_eq_gram` establishes the *actual equality of rational matrices*

$$R_n + \tfrac{3}{2} I = \tfrac{1}{m} J + \sum_{j=2}^{m} w_j B_j B_j^{\mathsf{T}}, \quad m = \lfloor n/2 \rfloor. \tag{15}$$

This result incorporates both parity branches, wraparound, all off-diagonal distances and the diagonal normalization. A separate Lean proof shows $1 \le d(u,v) \le m$ whenever $u \ne v$. The Stage 6 build completed 7,359 jobs, with 113 declarations passing the trusted-foundation axiom audit.

Stage 7: kernel-certified matrix positivity. The module `SpectralPSD.lean` uses the exact Stage 6 matrix theorem and proves, for every $n \ge 4$, that the *rational* matrix $R_n + \frac{3}{2} I$ is positive semidefinite in the formal `Matrix.PosSemidef` sense. The kernel checks nonnegative piecewise weights, the constant rank-one component, each actual incidence Gram product, and their finite weighted sum. Thus

$$x^{\mathsf{T}}(R_n + \tfrac{3}{2} I)x \ge 0 \qquad (x \in \mathbb{Q}^n). \tag{16}$$

The Stage 7 build passed 7,360 Lean jobs and audited 125 declarations within the standard foundational axiom allowlist. The conversion to a real-matrix/eigenvalue statement is a separate formal gate; the analytic consequence for real symmetric matrices remains available by the usual density and spectral arguments.

Stage 8: real Gram identity and spectral positivity. Two additional formal modules, `RealGram.lean` and `RealSpectral.lean`, transport the actual weighted cyclic-incidence Gram construction from $\mathbb{Q}$ to $\mathbb{R}$. The kernel verifies both the equality with the real shifted reciprocal-distance matrix and real positive semidefiniteness. It then proves

$$\operatorname{spec}_{\mathbb{R}}(R_n + \tfrac{3}{2} I) \subseteq [0, \infty) \qquad (n \ge 4). \tag{17}$$

Consequently the familiar lower bound $\lambda_{\min}(R_n) \ge -3/2$ follows via the standard scalar spectral shift; that last unshifted-eigenvalue formulation has not yet been compiled as its own Lean theorem. Stage 8 GitHub Actions passed 7,362 build jobs; 142 declarations survived the axiom audit with only the standard foundational allowlist.

Stage 9: parity-uniform rational comparisons. In `HarmonicComparison.lean`, Lean certifies exact harmonic-number recurrences and proves both rational spectral-spread upper bounds decrease strictly with the cycle half-length. Writing $q_o(m) = (4H_m + 3)/m$ and $q_e(m) = 4(2H_{m-1} + 1/m + 3/2)/(2m-1)$, the kernel proves

$$\begin{aligned} q_o(m) &< 2 - (22/105)^2 \qquad (m \ge 7), \\ q_e(m) &< 2 - (11/28)^2 \qquad (m \ge 8). \end{aligned} \tag{18}$$

The boundary evaluations $q_o(7) = 468/245$ and $q_e(8) = 1907/1050$ are exact and are joined to the universal monotonicity results, not inferred from finite samples. The Stage 9 kernel run passed 7,363 build jobs and audited 180 declarations. The precise trigonometric/adjacency eigenvalue comparison and finite $n = 3, \ldots, 14$ complementary cases still require Lean treatment.

Stage 10: trigonometric and all-cycle closure. The modules `TrigonometricEndgame.lean` and `AllCycleAnalytic.lean` use the mathlib inequality $\cos x \ge 1 - x^2/2$ and $\pi < 22/7$ to verify, without numerical approximation,

$$\begin{aligned} q_o(m) &< 2\cos\frac{\pi}{2m+1} \qquad (m \ge 7), \\ q_e(m) &< 2\cos\frac{\pi}{m} \qquad (m \ge 8). \end{aligned} \tag{19}$$

A further single Lean theorem selects the appropriate parity and proves the closed-form strict comparison for every $n \ge 15$. The accepted Stage 10 run completed 7,365 build jobs and audited 197 declarations. The actual cycle adjacency matrix over $\mathbb{Z}/n\mathbb{Z}$ is now defined in `AdjacencyDefinition.lean`, with its symmetry and entry nonnegativity certified in the 204-declaration kernel audit. This isolates the remaining ordered-eigenvalue theorem. This formally closes the *analytical* comparison, but it does not identify the right-hand cosine with the *ordered* second-smallest adjacency eigenvalue inside Lean.

Stage 11: final graph statement and exact bridge. The module `FinalTheoremProbe.lean` now defines the *actual* cycle gravity matrix by the published $4/((n-1)d(u,v))$ entries and its numerical range as the supremum minus infimum of the real spectrum. It also

uses mathlib's *ordered* eigenvalues (including multiplicity) of the actual cycle adjacency matrix, selecting the second-smallest at index $n-2$. The companion module `ActualGravityBridge.lean` kernel-proves

$$R_n + \tfrac{3}{2}I = \texttt{realShiftedReciprocalCycle}(n), \quad (20)$$

with R_n now defined by its zero-diagonal graph entries, and certifies the positive semidefiniteness of this *actual* shifted graph matrix. It also verifies the gravity scale factor $4/(n-1)$.

The Stage 11 Lean run passed 7,368 build jobs with 223 declarations admitted by the standard foundational axiom audit. The end-to-end numerical Graffiti inequality is accepted by the kernel *only conditionally*, assuming (i) identification of the actual second-smallest cycle adjacency eigenvalue with (8), and (ii) the actual gravity spectral-span bound (7). The theorem's statement therefore exposes rather than closes both missing spectral bridges. The finite converse $C_3, \ldots, C_{14}$ is also pending kernel certification. There is no claim of a complete unconditional Lean proof.

Stage 12: distance-shell upper matrix certificate. The modules `DistanceShellLaplacian.lean`, `DistanceShellAdjacency.lean`, and `DistanceShellDegreeBridge.lean` construct an independent positive-semidefinite *upper* matrix certificate, complementary to the shifted-Gram lower certificate. For each distance $1 \le d \le m$, let E_d be the undirected distance-d shell graph of C_n. Writing $L(E_d)$ for its ordinary graph Laplacian and D_n for the weighted diagonal shell-degree matrix, the kernel proves the exact identities

$$R_n = \sum_{d=1}^{m} \frac{1}{d} A(E_d),$$
$$L_n = \sum_{d=1}^{m} \frac{1}{d} L(E_d) = D_n - R_n. \quad (21)$$

Each graph Laplacian is positive semidefinite, so

$$D_n - R_n \succeq 0, \qquad \frac{4}{n-1} D_n - \mathrm{Gr}_n \succeq 0 \quad (n \ge 4). \quad (22)$$

These are *actual* cycle-distance matrices and the original gravity scaling, with no spectral target substituted into the definition. The Stage 12 kernel audit accepted 250 project declarations after 7,371 successful build jobs. The remaining graph-specific step is to prove inside Lean that $D_n = s_n I$, with $s_n = 2H_m$ for odd n and $s_n = 2H_{m-1} + 1/m$ for even n. Until that identity and the spectral extraction compile, (22) is not by itself the full numerical-span inequality.

Originality screen and claim calibration. The primary Roucairol–Cazenave source already refutes Graffiti 197 using C_{17} and conjectures only the C_{4r+1} family; the present claimed advance is the stronger sharp classification and constructive matrix certificate, not the first refutation. Earlier work also studies reciprocal-distance (Harary) spectra and their spreads. The focused source screen found no identical all-n threshold proof, but no search result establishes priority or substitutes for external specialist review. See the repository's prior-art audit.

Independent verification. GitHub Actions run 39131413230 passed 7,358 Lean build jobs. Its axiom audit accepted 100 declarations with only `propext`, `Classical.choice`, `Quot.sound`; the proof-placeholder check passed. Exact `fractions.Fraction` arithmetic independently confirmed all 7,259 cases $2 \le m \le 120$, $1 \le d \le m$. See the versioned proof ledger.

Claim boundary. Stages 1–12 now kernel-certify the exact shifted reciprocal Gram decomposition and its real spectral positivity, the odd/even harmonic and trigonometric analytic comparisons, and the actual gravity upper Loewner bound (22). The strict Graffiti inequality has been compiled only *conditionally* upon its actual spectral interfaces. The outstanding Lean proofs are: scalar shell-degree/row-sum normalization, gravity spectral-span extraction, the adjacency matrix's ordered second-smallest eigenvalue and multiplicities, and the finite nonviolating cases $C_3, \ldots, C_{14}$. The full sharp-cycle theorem therefore remains a candidate for end-to-end kernel certification and independent expert review.

6 Reproducibility and claim boundaries

The paper's proof is self-contained apart from classical elementary facts ($\pi < 22/7$, $\cos x \ge 1 - x^2/2$, and the cycle adjacency spectrum). The accompanying `verify.py` provides four checks: rational matrix-identity entries for orders 4 to 40, rational bounds for all small orders 3 to 14, rational evaluation of both boundary inequalities, and an independent floating-point Fourier cross-check through order 160. The numerical cross-check is deliberately *not* an ingredient of the theorem. Proof-critical component lemmas have now been independently kernel-checked in Lean 4/mathlib as documented in Section 5; the complete graph-theoretic theorem has not been formally certified.

Graffiti 197 itself was already refuted by Roucairol and Cazenave [5]; its resolution is not claimed as a new discovery. The contribution under review is the uniform analytic infinite-family theorem, the extended threshold, and the exact Gram decomposition. The targeted literature audit did not identify an earlier proof of this specific strengthening, but that negative search result is not a priority certificate. Mathematical validity, originality, and suitability for publication remain open to independent assessment.

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